

Education

University of Illinois Chicago, College of Engineering, Computer Science

Master of Science (MS)

GPA: 4.0

Indian Institute of Information Technology, Una, School of Computing, Computer Science and Engineering

Bachelor of Technology (B.Tech)

CGPA: 8.58/10

Work Experience

University of Illinois at Chicago

Graduate Hourly, Chicago, IL

Sept 2025 - present

- Optimizing the Auto-Ext project, exploring opportunities for optimal configuration and better scheduling policies.
- Preparing the project for formal publication, highlighting novel contributions to sched_ext and workload-adaptive scheduling research.

Optum Global Solutions, UnitedHealth Group

Software Engineer, India

Jul 2021 - Jul 2024

- Built a secure FHIR API server and application suite (Spring Boot and Maven) achieving **<40ms p95 latency** and **>99% uptime**, ensuring robust PII/PHI data handling.
- Led the migration of applications to the Kubernetes platform, setting up multiple environments while maintaining >99% application availability, resulting in enhanced scalability.
- Led the initiative to resolve multiple security vulnerabilities across codebases, including fixing a critical zero-day within 12 hours of discovery.
- Optimized data validation applications, achieving a **50% efficiency boost** and **100% increase in operating window**.
- Received the Bravo recognition award for spearheading vulnerability removal efforts.
- Coordinated a major database server upgrade with cross-functional teams, ensuring minimal disruption.

Projects

OAuth 2.0 Authorization Server: FastAPI, Python, JWT, PKCE

- Designed and implemented a fully RFC-compliant OAuth 2.0 Authorization Server supporting Authorization Code + PKCE flow, RS256-signed JWTs, token Introspection & Revocation, and JWKS public key discovery.
- Engineered modular FastAPI micro-services for user/client registration, secure token issuance and verification aligning with RFC 6749, 7636, 7519, 7662, 7009 & 8414.

Auto-Ext: Dynamic eBPF-powered workload profiler & scheduler dispatcher (*Work in Progress*)

- Developed system profilers to identify workloads in real time using eBPF and then use Linux's sched_ext (SCX) framework to automatically dispatch the optimal scheduler policy under varying load.
- Implemented the profilers in C, dispatcher in Python, plus orchestrations using shell scripts
- Ensured portability across kernels ≥ 6.12 or with sched_ext/eBPF support.
- Ongoing work focuses on reducing overhead and optimizing scheduling policies for mixed workloads.

CustOS: A Lightweight Custom Operating System

- Built a custom Linux-based OS under **250 MB**, achieving a 98% reduction to typical modern OS sizes (~15 GB).
- Integrated GUI, networking, and package management for extensibility and user customization.

Skills

Programming: C, C++, Java, Python, Solidity, SQL**Backend & Cloud:** Spring, J2EE, REST APIs, Docker, Kubernetes, Openshift, Azure**Testing & DevOps:** Jenkins, JUnit 5, SonarQube, Postman, Git**Systems & Theory:** Algorithm Design, Data Structures, Networking & Security**Creative Tools:** UI/UX Design, Figma, Adobe CC (Photoshop, Illustrator, Lightroom, Premiere Pro,), Canva

Positions of Responsibility

Coordinator, Team Pixonoids Led media & content creation for official college and club activities; managed graphics team for design and visual strategy.